CHAPTER 19 · DERIVATIONS

ALTERNATING CURRENT CIRCUITS

5 derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 PURE RESISTOR: CURRENT AND POWER

SYLLABUS 19.1(C), (E)

RESULT $I = I_0 \sin \omega t$ (in phase); $\langle P \rangle = \frac{1}{2}V_0 I_0 = I_{rms}^2 R$

1. $V = V_0 \sin \omega t$ across R: $I = \frac{V}{R} = \frac{V_0}{R} \sin \omega t = \mathbf{I_0 \sin \omega t}$, in phase with V.
2. Instantaneous power $P = VI = V_0 I_0 \sin^2 \omega t = \frac{1}{2}V_0 I_0 (1 - \cos 2\omega t)$.
3. The mean of $\cos 2\omega t$ over a cycle is zero: $\mathbf{\langle P \rangle = \frac{1}{2}V_0 I_0 = I_{rms} V_{rms} = I_{rms}^2 R}$.

2 PURE INDUCTOR: CURRENT, REACTANCE AND POWER

SYLLABUS 19.2(A), (D), (E)

RESULT $I = I_0 \sin(\omega t - \frac{\pi}{2})$; $X_L = \omega L$; $\langle P \rangle = 0$

1. $V = V_0 \sin \omega t$ is balanced by the back e.m.f.: $V_0 \sin \omega t = L \frac{dI}{dt}$.
2. Integrate: $I = -\frac{V_0}{\omega L} \cos \omega t = \mathbf{I_0 \sin(\omega t - \frac{\pi}{2})}$: the current lags the voltage by $\frac{\pi}{2}$.
3. $X_L = \frac{V_0}{I_0} = \mathbf{\omega L}$.
4. $P = VI = -V_0 I_0 \sin \omega t \cos \omega t = -\frac{1}{2}V_0 I_0 \sin 2\omega t$; its mean over a cycle is **zero**.

3 PURE CAPACITOR: CURRENT, REACTANCE AND POWER

SYLLABUS 19.3(A), (D), (E)

RESULT $I = I_0 \sin(\omega t + \frac{\pi}{2})$; $X_C = \frac{1}{\omega C}$; $\langle P \rangle = 0$

1. $Q = CV = CV_0 \sin \omega t$.
2. $I = \frac{dQ}{dt} = \omega C V_0 \cos \omega t = \mathbf{I_0 \sin(\omega t + \frac{\pi}{2})}$: the current leads the voltage by $\frac{\pi}{2}$.
3. $X_C = \frac{V_0}{I_0} = \mathbf{\frac{1}{\omega C}}$.
4. $P = VI = \frac{1}{2}V_0 I_0 \sin 2\omega t$; its mean over a cycle is **zero**.

4 RESONANCE FREQUENCY OF AN R-L-C CIRCUIT

SYLLABUS 19.4(F)

RESULT $f_0 = 1/(2\pi\sqrt{LC})$

1. $Z = \sqrt{R^2 + (X_L - X_C)^2}$ is smallest ($Z = R$) when $X_L = X_C$.
2. $\omega_0 L = \dfrac{1}{\omega_0 C}$, so $\omega_0 = \dfrac{1}{\sqrt{LC}}$.
3. **$f_0 = 1/(2\pi\sqrt{LC})$**; the current is then maximum and in phase with the supply.

5 IMPEDANCE OF A SERIES R-L-C CIRCUIT (PHASOR METHOD)

SYLLABUS 19.4

RESULT $Z = \sqrt{R^2 + (X_L - X_C)^2}$

1. The same current I flows through R, L and C. Draw the current phasor along the reference axis.
2. $V_R = IR$ is in phase with I; $V_L = IX_L$ leads I by 90°; $V_C = IX_C$ lags I by 90°.
3. Add the phasors: $V^2 = V_R^2 + (V_L - V_C)^2 = I^2[R^2 + (X_L - X_C)^2]$.
4. **$Z = \dfrac{V}{I} = \sqrt{R^2 + (X_L - X_C)^2}$**, and the phase angle is $\tan\varphi = \dfrac{X_L - X_C}{R}$.
5. For R-C only: $Z = \sqrt{R^2 + X_C^2}$; for R-L only: $Z = \sqrt{R^2 + X_L^2}$.